

MUTUAL FUND ANALYTICS

End-to-End Financial Data Analytics Platform

ETL • SQLite • SQL • EDA • Performance Analytics • Streamlit Dashboard • Automated Reporting

Bluestock Fintech Capstone Project

Final Project Report — Academic, Portfolio & Viva Documentation

Prepared for MCA academic submission, internship evaluation and Data Analyst portfolio review

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1. Executive Summary

Mutual Fund Analytics is an end-to-end financial data analytics platform built as a Bluestock Fintech capstone project. It connects data ingestion and validation with a reusable SQLite analytical layer, exploratory analysis, fund performance and risk analytics, an interactive Streamlit dashboard, weekly reporting and automated email delivery. The workflow is designed as a repeatable system: Extract → Transform → Load → Analyse → Visualise → Report → Automate.

The validated fund universe contains 40 schemes across large-cap, mid-cap, small-cap, flexi-cap, hybrid, gilt, liquid and other categories. AMFI-code validation matched 40 fund-master codes to 40 NAV-history codes with zero missing codes, confirming referential integrity between the fund master and the historical NAV dataset. Historical analytics are kept distinct from selected live-NAV retrieval through the mfapi.in public API, which supports the project's recurring reporting workflow.

40 / 40	≈ 46,000	41.1%	₹31,002 Cr
Validated Schemes (0 missing codes)	Historical NAV Records 2022-01-03 to 2026-05-29	Investor Segment Age 26-35	All-time-high Monthly SIP Inflow
66.5% / 33.5%	13.26 → 26.12 Cr	₹12.5 L Cr	4
Investor Gender Split Male / Female	Total Folio Growth Jan 2022 – Dec 2025	Leading Fund House AUM (SBI Mutual Fund, latest)	Dashboard Modules Industry / Fund / Investor / SIP

The report that follows documents the project's business rationale, architecture, ETL pipeline, database design, analytics (exploratory, performance and advanced), dashboard, automation and email reporting, together with key findings, business insights, limitations and future scope.

2. Project Overview

Project Name: Mutual Fund Analytics • Project Type: End-to-End Data Analytics / ETL / SQL / Database / Dashboard / Automated Reporting Project • Context: Bluestock Fintech Capstone Project.

The project builds a mutual fund analytics system that collects fund data, cleans and transforms it, loads processed data into SQLite, performs SQL-based and exploratory analysis, calculates performance metrics, creates financial visualisations, provides an interactive dashboard, generates automated weekly reports, sends automated email summaries, and runs the ETL/reporting workflow through Windows Task Scheduler.

The project analyses approximately 40 mutual fund schemes, with historical NAV data covering 2022-01-03 to 2026-05-29, and supporting datasets (AUM, SIP, folios, category inflows, investor transactions, scheme performance and portfolio holdings) spanning comparable periods.

3. Problem Statement

Mutual fund information in practice is fragmented: NAV history, AUM by fund house, SIP inflows, category-level flows, investor demographics and scheme-level performance metrics typically arrive in different formats and structures. Before this data can support analysis or decision-making, it must be extracted, validated, cleaned, linked by common identifiers (such as

AMFI scheme codes) and stored in a queryable analytical layer.

Without a consolidated system, an analyst would need to repeat this cleaning and joining process for every question asked of the data, and would have no repeatable way to track how metrics change over time. This project addresses that gap by building a single analytical pipeline and dashboard that convert raw, fragmented mutual-fund data into a consistent, explorable and continuously refreshable analytical resource.

4. Objectives

- Build a reliable ETL pipeline covering extraction, cleaning, validation and loading of mutual-fund data.
- Validate fund identifiers (AMFI codes) and confirm historical NAV coverage across the full fund universe.
- Store processed analytical data in a structured SQLite database using a dimension/fact design.
- Perform exploratory data analysis across NAV, AUM, SIP, category and investor dimensions.
- Calculate return, benchmark-relative and risk metrics (CAGR, Sharpe, Sortino, Alpha, Beta, volatility, maximum drawdown) for the fund universe.
- Provide an interactive Streamlit dashboard covering industry, fund performance, investor and SIP/market views.
- Automate live NAV updates, weekly report generation and email delivery using a scheduled pipeline.

5. Business Use Case

The platform is designed around the kind of questions a mutual-fund analyst, distributor or internal research team would routinely ask: Which fund houses are growing AUM fastest? Which schemes combine strong 3-year returns with acceptable risk? Are SIP inflows accelerating or slowing, and in which categories? What does the investor base look like by age, gender and city tier? Consolidating these views in one SQLite-backed dashboard removes the need to rebuild the same joins and calculations for every new question, and the automated weekly report keeps a standing summary current without manual intervention.

The platform is analytical and educational in nature. It does not place trades, does not manage investor money and does not constitute investment advice; see Section 23 (Limitations) and the disclaimer on the final page.

6. Data Sources

The project combines supplied project datasets with public mutual-fund reference information. AMFI-related identifiers and the mfapi.in public NAV API support NAV-related data, while AUM, SIP, folio, investor-transaction and scheme-performance datasets support the broader analytics.

Source	Role in the Project
Fund master (dim_fund)	Reference list of 40 validated schemes with AMFI codes, fund house and category.
Historical NAV history	~46,000 daily NAV observations, 2022-01-03 to 2026-05-29, used for return and risk calculations.
mfapi.in (live NAV)	Selected live NAV retrieval used by the automated reporting workflow.
AUM by fund house	Quarterly AUM observations across 10 fund houses, 2022-2025.

Source	Role in the Project
Monthly SIP inflows	48 monthly observations of SIP inflow, active accounts and SIP AUM.
Category inflows	Monthly net inflow by fund category (12 categories).
Folio counts	Quarterly total/equity/debt/hybrid/other folio counts, 2022-2025.
Scheme performance	Scheme-level return, benchmark, risk and rating metrics for the 40-scheme universe.
Portfolio holdings	Stock-level holdings (sector, weight, market value) supporting sector-allocation analysis.
Benchmark indices	Daily closing values for NIFTY50, NIFTY100, NIFTY500, NIFTY Midcap150, BSE Smallcap, CRISIL Liquid and CRISIL Gilt, 2022-2026.
Investor transactions	Synthetic investor-level dataset supporting demographic and geographic analysis.

7. Dataset Description

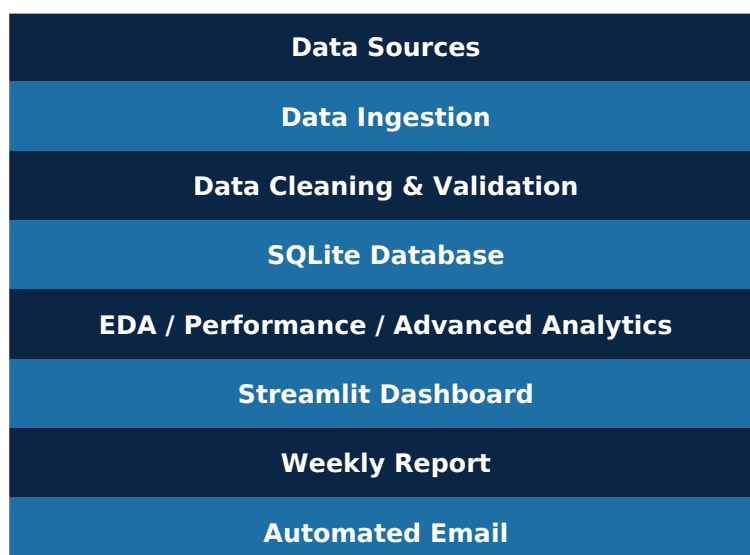
The datasets underpinning the analysis are summarised below. Record counts reflect the supplied processed extracts used to build this report.

Dataset	Records	Key Columns
Fund master	40 schemes	amfi_code, scheme_name, fund_house, category, plan
Scheme performance	40 rows	return_1yr/3yr/5yr_pct, alpha, beta, sharpe_ratio, sortino_ratio, std_dev_ann_pct, max_drawdown_pct, aum_crore, expense_ratio_pct, morningstar_rating, risk_grade
Fund scorecard	40 rows	final_rank, CAGR_3Y, sharpe_ratio, alpha, expense_ratio_pct, max_drawdown, fund_score, risk_grade
VaR / CVaR report	40 rows	VaR_95, CVaR_95 (absolute and %)
Sector HHI report	34 rows	sector_hhi, category, concentration
Portfolio holdings	322 rows	amfi_code, stock_symbol, sector, weight_pct, market_value_cr, portfolio_date
AUM by fund house	90 rows	date, fund_house, aum_lakh_crore, num_schemes
Monthly SIP inflows	48 rows	month, sip_inflow_crore, active_sip_accounts_crore, yoy_growth_pct
Category inflows	144 rows	month, category, net_inflow_crore
Folio counts	21 rows	month, total/equity/debt/hybrid/others_folios_crore
Benchmark indices	8,050 rows	date, index_name, close_value

Investor transaction data used for demographic analysis (age, gender, city tier) is synthetic and generated for educational analysis; it should not be generalised beyond the supplied project dataset.

8. System Architecture

The architecture follows a practical data-engineering pattern that carries data from source files through to business-facing reporting. The master pipeline is orchestrated by `run_pipeline.py`.



Extract: historical NAV, fund master, AUM, SIP, category, folio and investor datasets, plus selected live NAV data from mfapi.in.

Transform: date/type cleaning, missing-value handling, validation, standardisation, dataset joins and analytical calculations (returns, CAGR, Sharpe, Sortino, Alpha, Beta, VaR/CVaR, HHI).

Load: processed datasets are loaded into the project's SQLite database (SQL/bluestoke_mf.db).

Analyse / Visualise: SQL analysis, EDA, performance and advanced analytics feed the Streamlit dashboard and the official project charts reproduced in this report.

Report / Automate: a weekly HTML report and a dynamic email summary are generated by the pipeline and triggered on a schedule via Windows Task Scheduler.

9. ETL Pipeline

The ETL workflow is implemented as a sequence of Python scripts orchestrated by a master pipeline script, executed via Python subprocess calls:

Script	Responsibility
data_ingestion.py	Extracts raw source data (historical NAV, fund master, AUM, SIP, category, folio, investor and performance datasets; selected live NAV via mfapi.in).
data_cleaning.py	Cleans and transforms extracted data: date parsing, type correction, missing-value handling, validation and standardisation.
load_database.py	Loads processed, validated datasets into the SQLite database using the dimension/fact schema.
run_pipeline.py	Master orchestrator; runs the ingestion, cleaning, loading, reporting and email steps in sequence via subprocess.
weekly_email_report.py	Generates the weekly HTML report and dynamic email summary from the current database state.

Running `run_pipeline.py` end-to-end refreshes live NAV data, updates the SQLite database, regenerates the weekly report and sends the email summary in one execution, which is what allows the workflow to be scheduled unattended.

10. Data Cleaning & Transformation

- **Date handling:** raw date fields are parsed into consistent datetime types across NAV, AUM, SIP, category, folio and benchmark datasets so that time-series joins and resampling are reliable.

- Type correction: numeric fields (NAV, AUM in crore/lakh crore, inflow amounts, percentages) are cast to numeric types, with locale-specific formatting normalised.
- Missing-value handling: incomplete or malformed records are identified during cleaning before being loaded into the fact tables.
- Validation: AMFI scheme codes present in the fund master are reconciled against the codes present in the NAV history to confirm referential integrity (see Section 12).
- Standardisation: scheme names, fund-house names and category labels are standardised so that grouping and filtering behave consistently across the dashboard and reporting layer.
- Joins and analytical calculations: cleaned datasets are joined on AMFI code / scheme name to compute downstream metrics such as returns, CAGR, Sharpe, Sortino, Alpha, Beta, VaR, CVaR and sector HHI.

11. Database Design

SQLite acts as the common analytical source for both the dashboard and the reporting workflow. The schema follows a dimension/fact pattern that separates static reference data from time-series and transactional observations.

Table	Type	Purpose
dim_fund	Dimension	Fund master / reference information: AMFI code, scheme name, fund house, category, plan.
fact_nav	Fact	Historical and updated NAV observations keyed by AMFI code and date.
fact_aum	Fact	AUM observations by fund house and date.
fact_performance	Fact	Scheme-level return and risk metrics (CAGR, Sharpe, Sortino, Alpha, Beta, drawdown).
fact_transactions	Fact	Investor transaction records supporting demographic and geographic analysis.
fact_sip	Fact	SIP inflow and industry-trend information.

Database Architecture

dim_fund (reference)	fact_nav / fact_aum / fact_sip	fact_performance	fact_transactions
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Dimension/fact schema: dim_fund is the shared reference table joined by AMFI code to each fact table.

12. SQL Analysis

SQL queries against the SQLite database support fund counts, fund-house summaries, AUM/NAV/SIP analysis, transaction analysis and the filtering used throughout the dashboard. A key validation query reconciles the fund master against the NAV history to confirm data-quality before analysis:

40	40	0
Fund-master AMFI codes	NAV-history AMFI codes	Missing codes

This 40/40/0 validation result confirms that every scheme in the fund master has corresponding NAV history, so downstream performance and risk calculations are not silently dropping schemes for lack of price data.

13. Exploratory Data Analysis

EDA converts cleaned records into business-oriented patterns across NAV, AUM, SIP, category and folio dimensions. The figures below are official project visualisations, reproduced from the project's generated charts and, where the underlying processed data was supplied, regenerated directly from that data for this report.

Daily NAV Trend of 40 Mutual Fund Schemes (2022–2026)

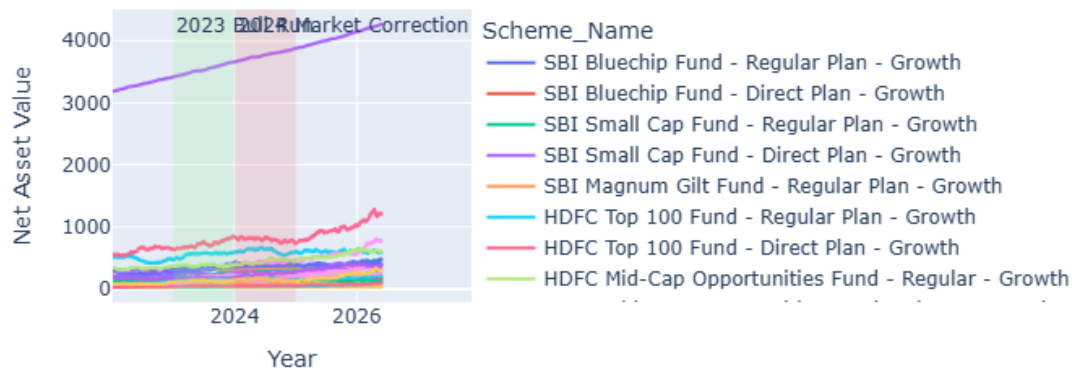


Figure 1: Daily NAV Trend of 40 Mutual Fund Schemes, 2022–2026 (official project chart, annotated with the 2023 mid-cycle rally and 2024 market-correction windows).

The NAV trend view plots all 40 schemes together, showing broad upward NAV movement across the period with a visible correction window in 2024 before recovery. Individual scheme trajectories diverge in magnitude by category (see Section 15 for scheme-level return and risk detail).

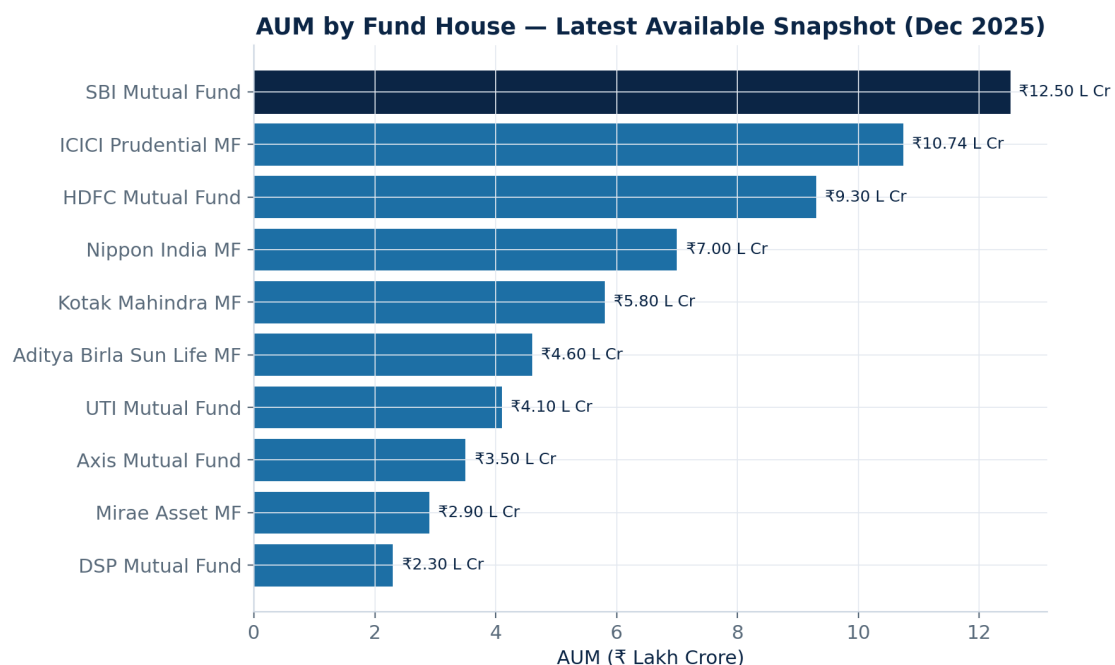


Figure 2: AUM by Fund House — Latest Available Snapshot (regenerated from 03_aum_by_fund_house.csv).

SBI Mutual Fund leads the ten displayed fund houses on AUM at the latest snapshot date, followed by ICICI Prudential MF and HDFC Mutual Fund. The AUM ranking is broadly stable across the fund houses shown, with a visible gap between the top three houses and the remaining seven.

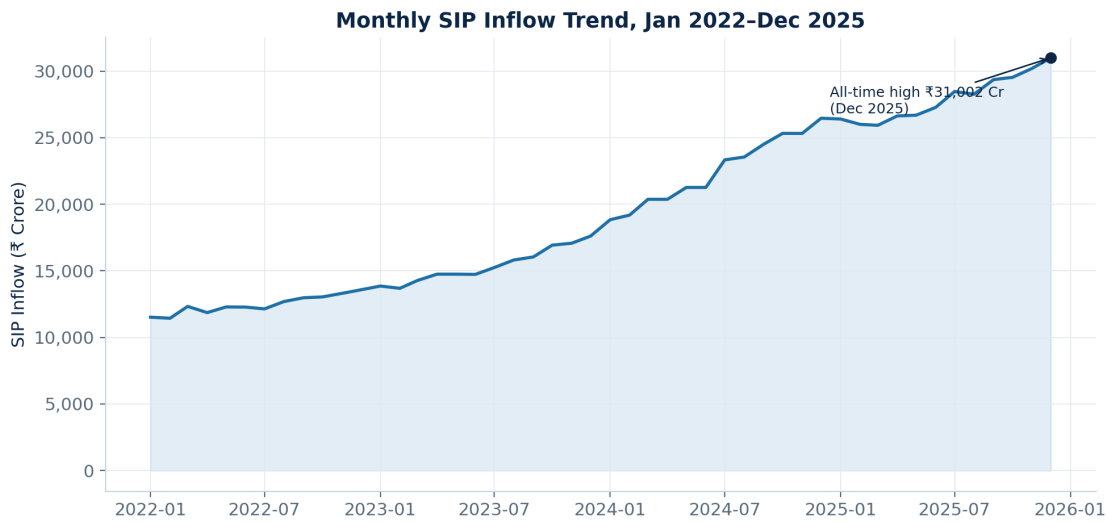


Figure 3: Monthly SIP Inflow Trend (regenerated from 04_monthly_sip_inflows.csv).

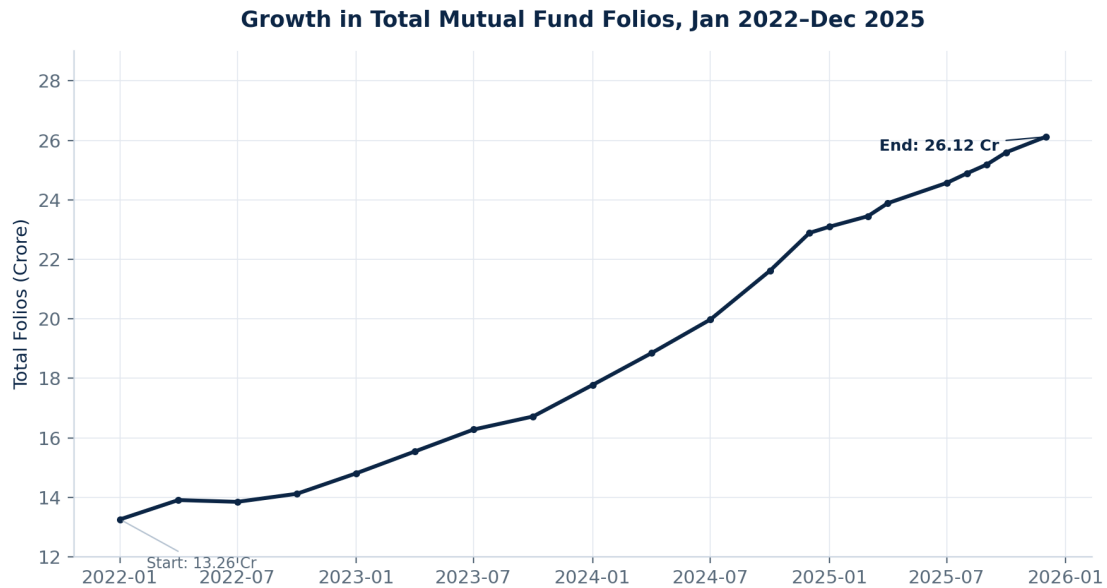


Figure 4: Growth in Total Mutual Fund Folios (regenerated from 06_industri_folio_count.csv).

Total folios nearly doubled across the observed period, indicating sustained growth in the number of distinct investor holdings tracked by the dataset.

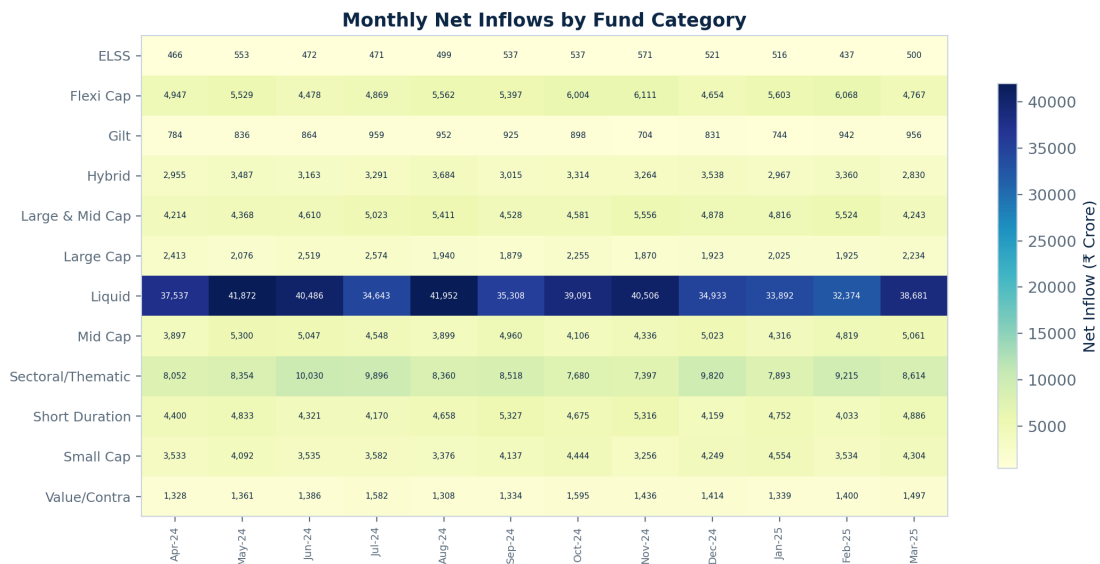


Figure 5: Monthly Net Inflows by Fund Category (regenerated from 05_category_inflows.csv).

The Liquid category consistently shows the largest monthly net inflows of the twelve categories tracked, reflecting its role as a short-term parking instrument, followed by Sectoral/Thematic and Flexi Cap categories.

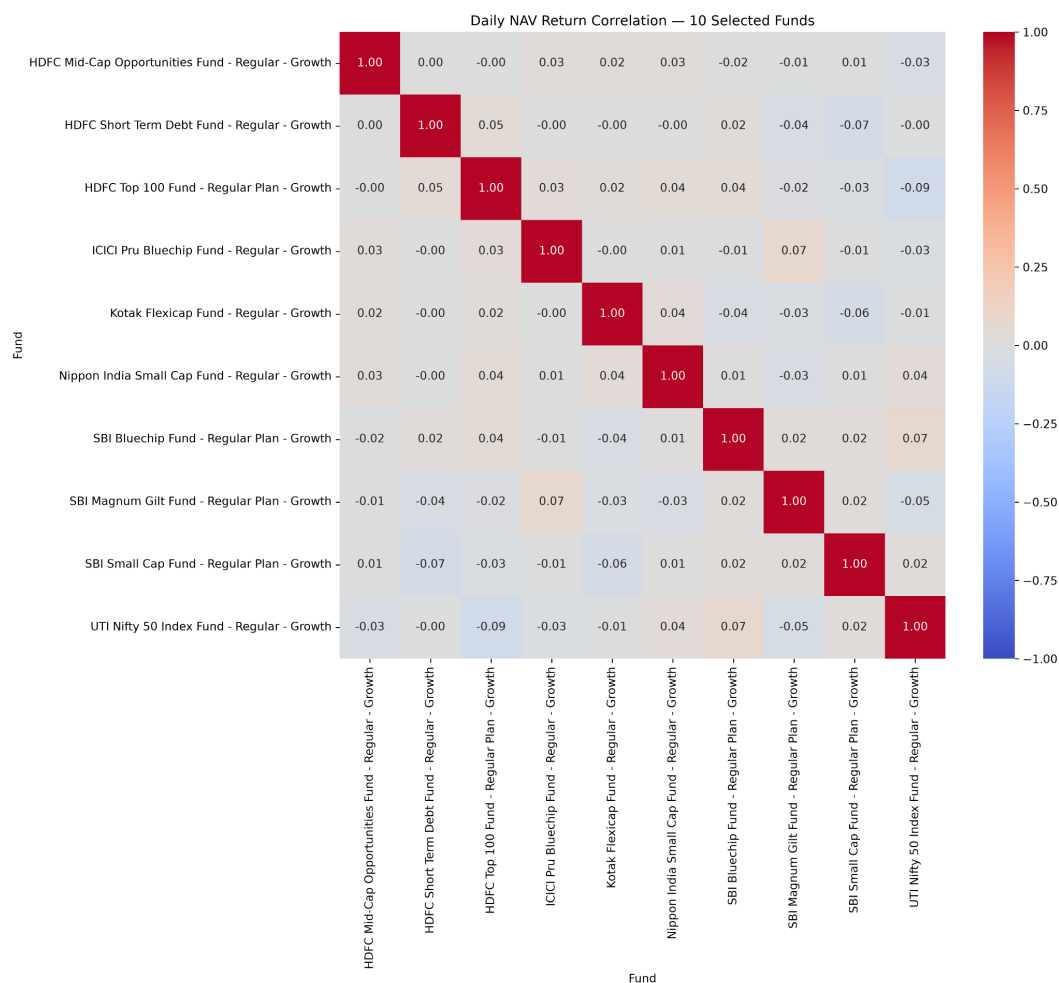


Figure 6: Daily NAV Return Correlation — 10 Selected Funds (official project chart).

Pairwise daily-return correlations among the 10 selected funds are generally low, which is consistent with the funds spanning different categories (large-cap, small-cap, gilt, flexi-cap)

rather than closely tracking a single market segment.

14. Mutual Fund Performance Analytics

The performance layer produces scheme-level records covering return, benchmark and risk measures for the full 40-scheme universe: daily and annualised returns, 1-year/3-year/5-year return where applicable, Sharpe ratio, Sortino ratio, Alpha, Beta, annualised volatility (standard deviation) and maximum drawdown.

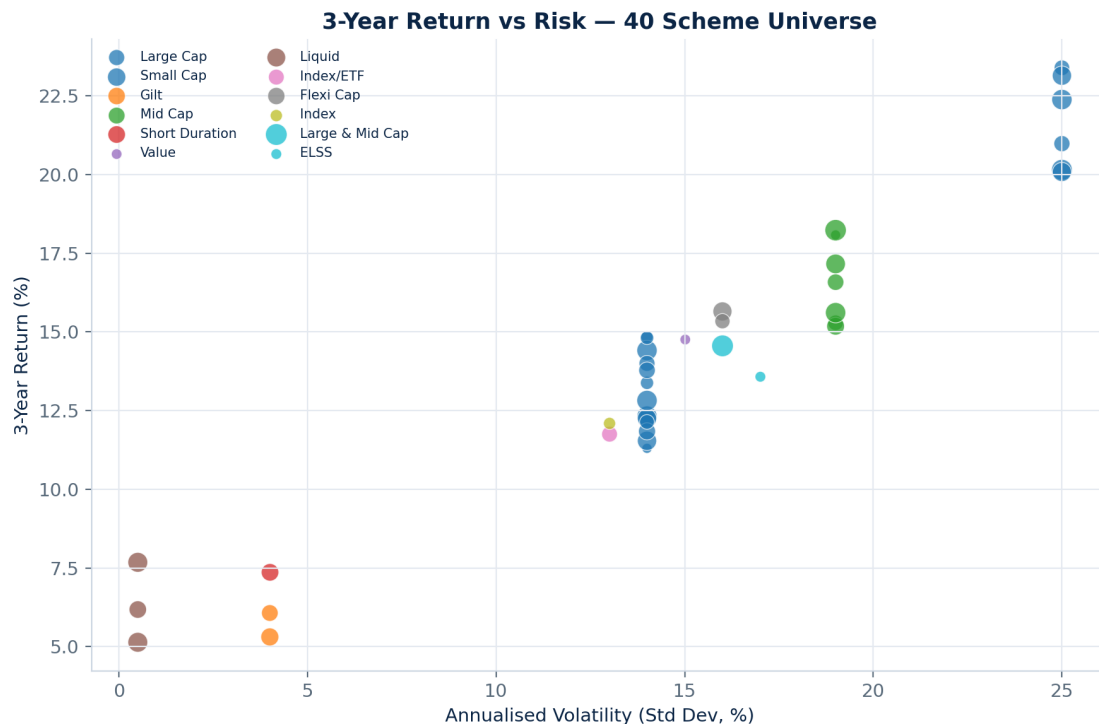


Figure 7: 3-Year Return vs Risk — 40 Scheme Universe, bubble size proportional to AUM (regenerated from 07_scheme_performance.csv).

Small-cap and mid-cap schemes cluster at higher volatility and higher 3-year return; gilt and large-cap schemes cluster at the lower-risk end. The pattern is consistent with expected category-level risk-return positioning.

Top 10 Schemes by 3-Year Return

Scheme	Category	1Y Ret. %	3Y Ret. %	Sharpe	Risk Grade
SBI Small Cap Fund Reg. - Growth	Small Cap	24.56	23.39	0.94	Very High
SBI Small Cap Fund Dir. - Growth	Small Cap	20.59	23.14	0.93	Very High
ABSL Small Cap Fund - Regular - Growth	Small Cap	24.93	22.38	0.90	Very High
Axis Small Cap Fund - Regular - Growth	Small Cap	21.97	20.98	0.84	Very High
Nippon India Small Cap Fund - Regular - Growth	Small Cap	21.30	20.15	0.81	Very High
DSP Small Cap Fund - Regular - Growth	Small Cap	20.20	20.08	0.80	Very High
Kotak Emerging Equity Fund - Regular - Growth	Mid Cap	17.12	18.23	0.96	High
ICICI Pru Midcap Fund - Regular - Growth	Mid Cap	14.02	18.08	0.95	High
DSP Midcap Fund - Regular - Growth	Mid Cap	14.12	17.16	0.90	High

Scheme	Category	1Y Ret. %	3Y Ret. %	Sharpe	Risk Grade
HDFC Mid-Cap Opportunities Fund - Regular - Growth	Mid Cap	15.43	16.58	0.87	High

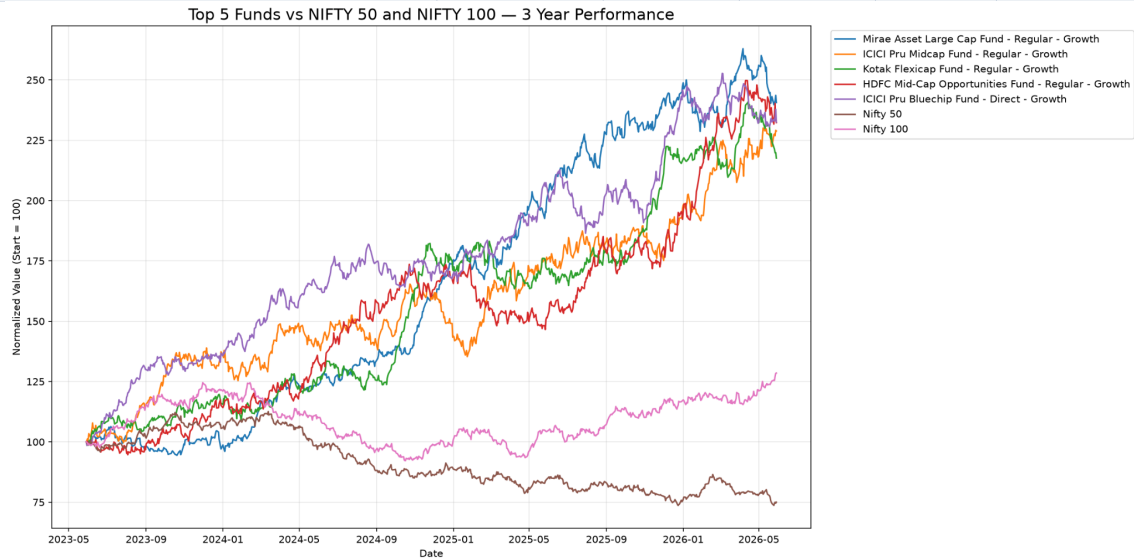


Figure 8: Top 5 Funds vs NIFTY 50 and NIFTY 100 — 3-Year Normalised Performance (official project chart).

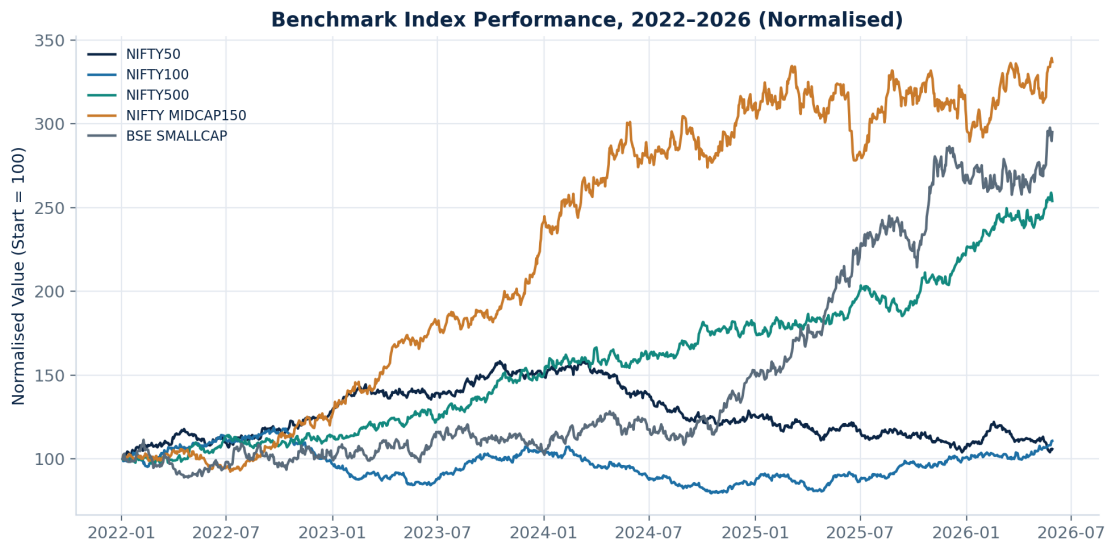


Figure 9: Benchmark Index Performance, 2022-2026, Normalised to 100 (regenerated from 10_benchmark_indices.csv).

The broader benchmark set (NIFTY 50, NIFTY 100, NIFTY 500, NIFTY Midcap 150 and BSE Smallcap) shows small/mid-cap indices delivering higher normalised growth than large-cap indices over the same window, which contextualises the fund-level return figures above.

15. Investor Analytics

Investor analysis covers age, gender, SIP behaviour and city-tier participation using the supplied synthetic investor-transaction dataset. These findings describe the analysed project dataset and should not be generalised to the full Indian mutual-fund investor population.

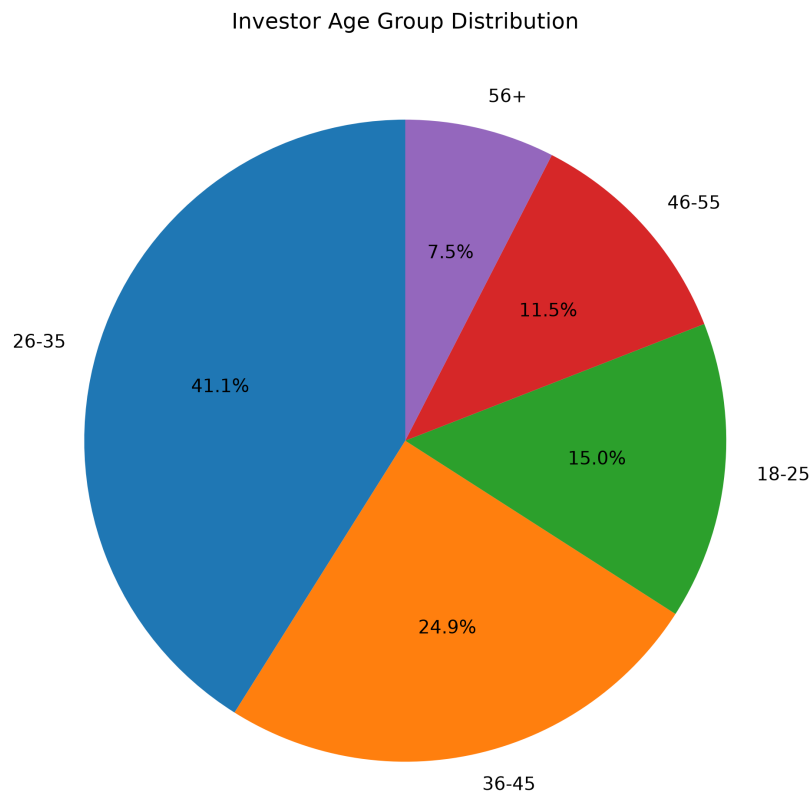


Figure 10: Investor Age Group Distribution (official project chart).

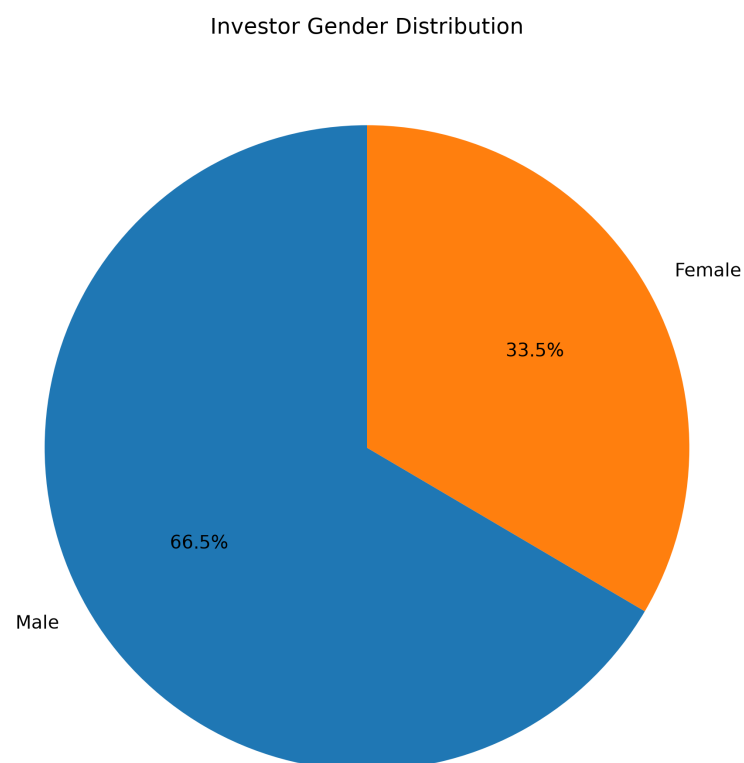


Figure 11: Investor Gender Distribution (official project chart).

- The 26–35 age band is the largest displayed investor segment at approximately 41.1%, followed by 36–45 at approximately 24.9%.

- Male investors represent approximately 66.5% of the analysed synthetic transaction dataset; female investors approximately 33.5%.
- T30 (top-30) cities contribute more total SIP investment than B30 (beyond top-30) cities in the analysed project data.

16. AUM & SIP Analytics

The AUM dataset covers Mar 2022 to Dec 2025 across 10 fund houses. SIP inflow data covers 48 monthly observations from Jan 2022 to Dec 2025.

Fund House	AUM (₹ Lakh Crore)	Schemes
SBI Mutual Fund	12.50	186
ICICI Prudential MF	10.74	216
HDFC Mutual Fund	9.30	195
Nippon India MF	7.00	177
Kotak Mahindra MF	5.80	168
Aditya Birla Sun Life MF	4.60	199
UTI Mutual Fund	4.10	142
Axis Mutual Fund	3.50	95
Mirae Asset MF	2.90	56
DSP Mutual Fund	2.30	88

Monthly SIP inflow grew from ₹11,517 crore at the start of the series to a peak of ₹31,002 crore, with year-on-year growth in the most recent observations in the mid-to-high teens (percentage).

17. Sector & Category Analysis

Category inflow analysis compares monthly net inflows across mutual-fund categories (Section 13, Figure 4). Sector allocation, shown below, is computed directly from the project's portfolio-holdings dataset (weighted by market value of each stock holding) and provides an aggregate view of equity-fund sector exposure across the fund universe.

Aggregate Sector Allocation Across Equity Fund Holdings

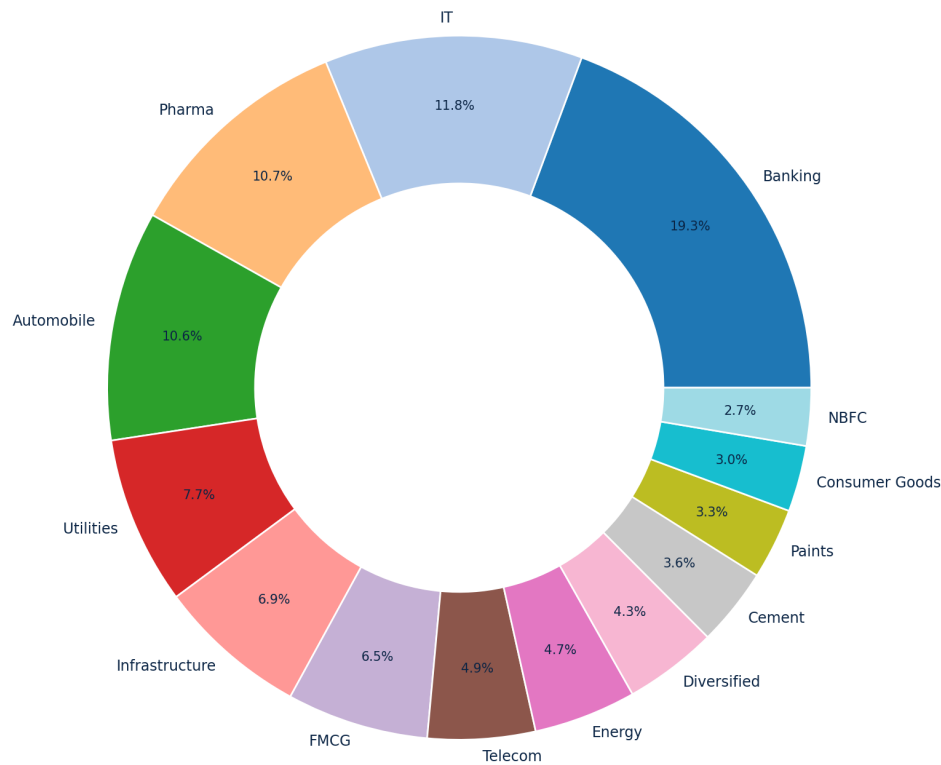


Figure 12: Aggregate Sector Allocation Across Equity Fund Holdings, weighted by market value (regenerated from 09_portfolio_holdings.csv).

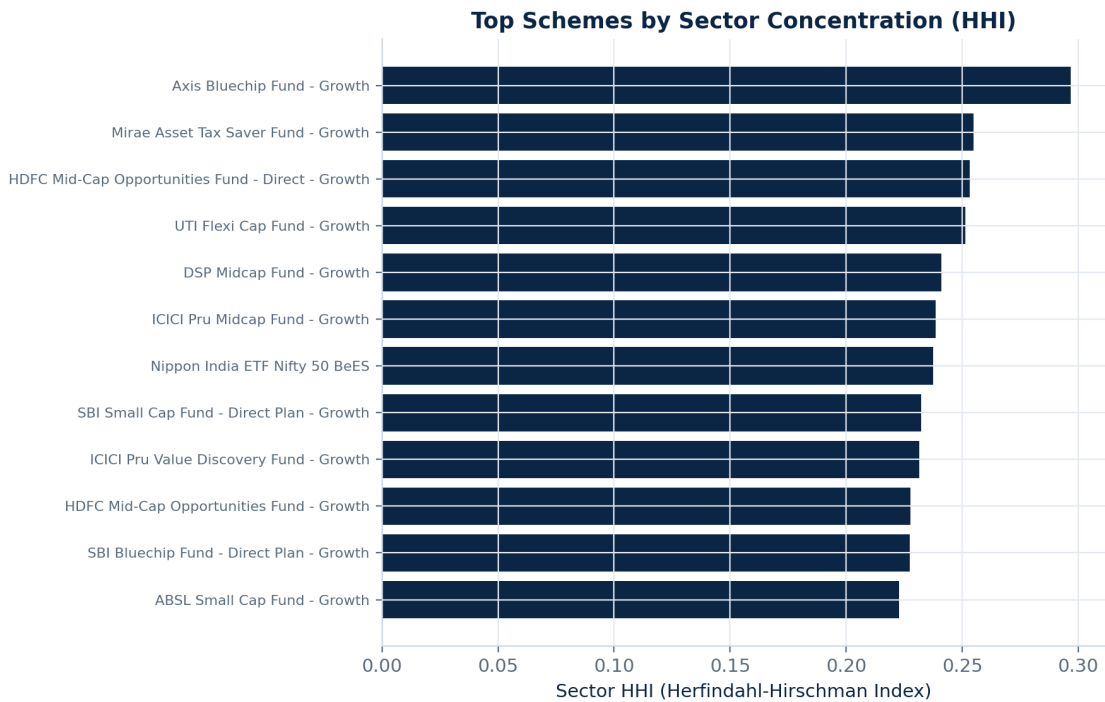


Figure 13: Top Schemes by Sector Concentration — Herfindahl-Hirschman Index (regenerated from sector_hhi_report.csv).

Banking, IT and Pharma are the largest sector exposures across the equity holdings analysed. The HHI view identifies which individual schemes carry the most concentrated sector exposure; schemes flagged “High Concentration” warrant closer diversification review than schemes with lower HHI values.

18. Advanced Analytics

Beyond descriptive EDA and standard performance metrics, the project includes risk-focused advanced analytics computed across the 40-scheme universe:

- Historical 95% VaR and CVaR — estimated daily downside risk and expected shortfall per scheme, computed from historical NAV returns.
- Rolling 90-day Sharpe — a rolling risk-adjusted return measure for tracking how a scheme's risk-return profile evolves over time.
- Investor cohort analysis — grouping investors by acquisition period / behaviour for retention-style analysis.
- SIP continuation / at-risk analysis — identifying SIP accounts at risk of discontinuation.
- Risk-grade based fund recommendation logic — a scoring approach (fund_score, final_rank) combining CAGR, Sharpe, Alpha, expense ratio and drawdown into a single ranked scorecard.
- HHI-based sector concentration — covered in Section 17.

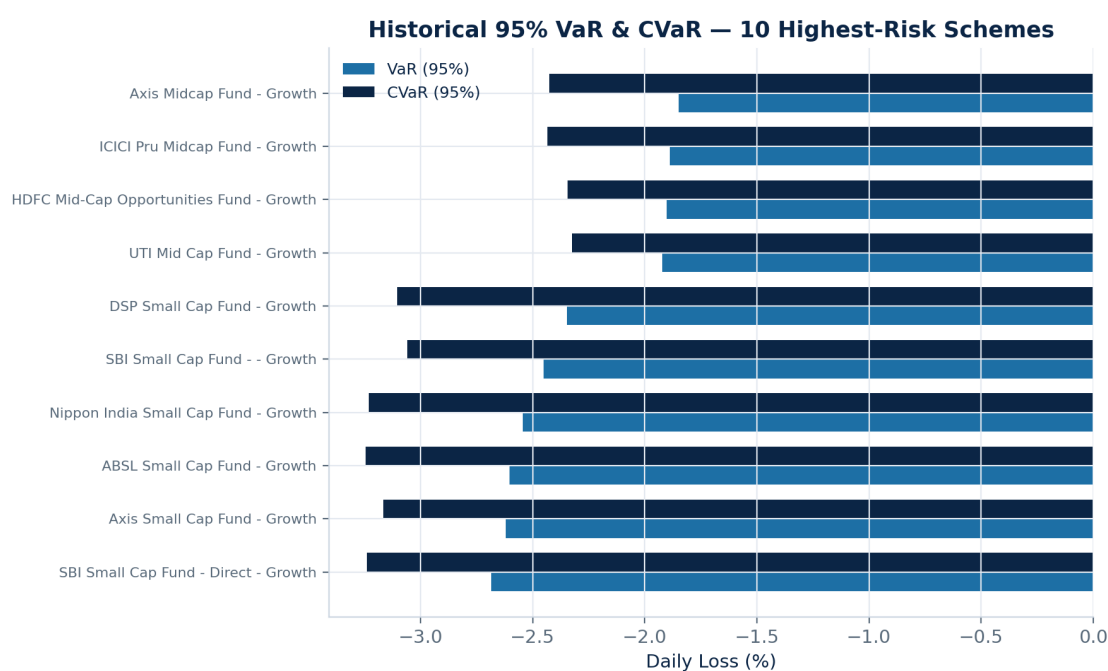


Figure 14: Historical 95% VaR & CVaR — 10 Highest-Risk Schemes (regenerated from var_cvar_report.csv).

Fund Performance Scorecard — Top 10 Ranked Schemes

Rank	Scheme	CAGR 3Y %	Sharpe	Fund Score	Risk Grade
1	Mirae Asset Large Cap Fund - Growth	34.0	1.45	86.2	Moderate
2	ICICI Pru Midcap Fund - Growth	31.8	1.18	82.2	High
3	Kotak Flexicap Fund - Growth	29.6	1.31	82.0	High
4	HDFC Mid-Cap Opportunities Fund - Growth	32.4	1.09	81.1	High

Rank	Scheme	CAGR 3Y %	Sharpe	Fund Score	Risk Grade
5	ICICI Pru Bluechip Fund - Direct - Growth	32.5	1.03	80.0	Moderate
6	Axis Midcap Fund - Growth	35.1	1.00	77.4	High
7	SBI Bluechip Fund Plan - Growth	30.5	1.21	75.0	Moderate
8	Mirae Asset Tax Saver Fund - Growth	29.2	1.23	73.9	High
9	ABSL Frontline Equity Fund - Growth	29.0	1.03	68.4	Moderate
10	SBI Small Cap Fund Plan - Growth	26.7	0.95	67.4	High

The scorecard is the same composite ranking exposed on the Fund Performance dashboard page (Section 19, Figure 12), combining three-year CAGR, Sharpe ratio, Alpha, expense ratio and maximum drawdown into a single fund_score used to rank schemes.

19. Dashboard

The current dashboard is a local Streamlit application backed by SQLite, comprising four pages: Industry Overview, Fund Performance, Investor Analytics and SIP & Market Trends. Filters and cached/on-demand loading support interactive exploration without loading unnecessary data at startup.

Industry Overview Dashboard



Figure 15: Streamlit Industry Overview Dashboard — KPI cards for Total AUM, Latest SIP Inflow, Latest Folios and Schemes in Dataset, with AUM trend, AUM by AMC, and Year/Fund-House filters.

This page answers industry-level business questions: How large is total tracked AUM, and how is it trending? Which fund houses hold the largest share? How many schemes and folios does the dataset currently cover? Year and fund-house filters let a user narrow the view to a specific period or AMC.

Fund Performance Dashboard

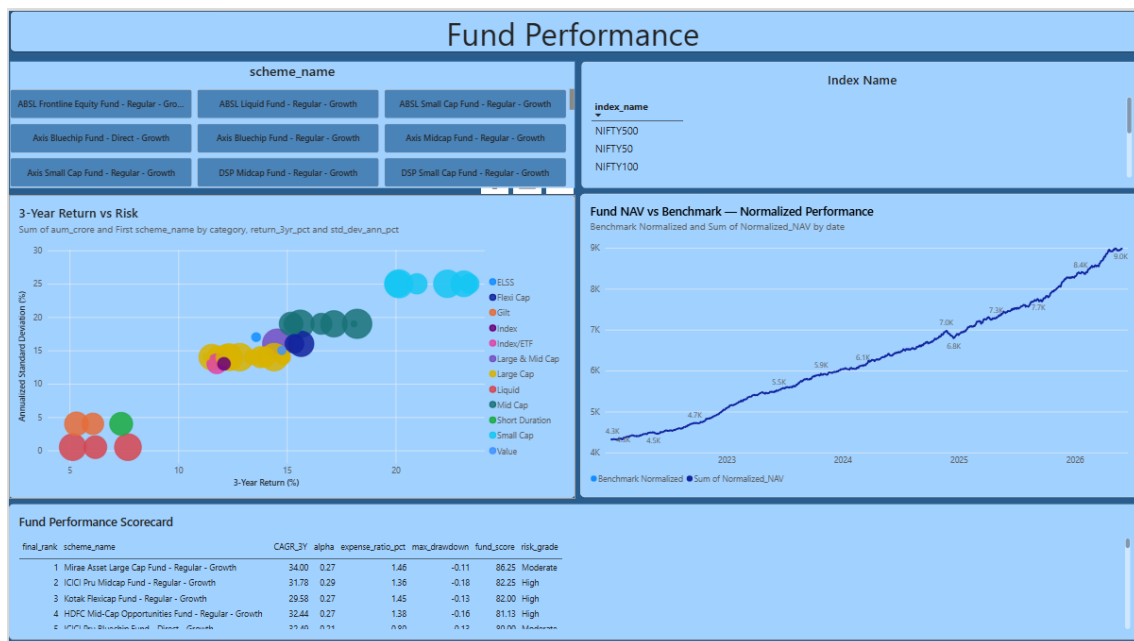


Figure 16: Streamlit Fund Performance Dashboard — fund/index selection, 3-Year Return vs Risk scatter, Fund NAV vs Benchmark line chart, and Fund Performance Scorecard table.

Users select a fund and a benchmark index (NIFTY 500 / NIFTY 50 / NIFTY 100) to compare normalised NAV against the chosen benchmark, examine the fund's position on the return-vs-risk scatter, and review its rank on the Fund Performance Scorecard (CAGR, Alpha, expense ratio, max drawdown, fund_score, risk grade).

Investor Analytics & SIP / Market Trends

The Investor Analytics page presents the age, gender and city-tier views documented in Section 15. The SIP & Market Trends page presents the monthly SIP inflow trend and category-level net inflows documented in Sections 13 and 16. Not available in the supplied project information: pixel-level screenshots of these two specific pages were not included among the supplied project evidence, so this report documents their content from the underlying data and from the other project materials rather than reproducing a screenshot.

20. Automation & Email Reporting

The automated workflow combines live NAV retrieval, database refresh, weekly report generation and email delivery in a single scheduled pipeline execution.

Step	Detail
Pipeline sequence	data_ingestion.py → data_cleaning.py → load_database.py → weekly report → email, orchestrated by run_pipeline.py via Python subprocess.
Scheduled task	Windows Task Scheduler task MutualFundAnalytics_ETL launches run_pipeline.py with the project root as the working directory.
Editor independence	VS Code does not need to remain open; the scheduled task runs the Python executable directly.
Power requirement	The Windows machine must be powered on and able to execute the scheduled task; if the machine is fully powered off, the task does not run unless Windows is separately configured to wake the computer for scheduled tasks.
Email delivery	Resend is used for email delivery. External-recipient delivery depends on the sender/domain configuration and Resend's account-level restrictions (see Section 23).
Email content design	The email focuses on a dynamic summary and KPIs rather than depending on chart images rendering correctly inside mobile email clients.

Observed successful automation test output: "Live NAV updated. SQLite database updated. Weekly report generated. Email sent successfully." This confirms the end-to-end pipeline executes without manual intervention when triggered.

Live NAV retrieval is implemented for selected schemes via mfapi.in, not as a continuously live feed across all 40 schemes; the distinction between historical NAV (used for performance/risk analytics) and selected live NAV (used to keep the automated report current) is maintained throughout the project.

21. Key Findings

- 40/40 AMFI-code validation between the fund master and NAV history, with zero missing codes.
- 26–35 is the largest displayed investor age segment at approximately 41.1% of the analysed synthetic transaction dataset.
- Male investors represent approximately 66.5% of the analysed dataset; female investors approximately 33.5%.
- Monthly SIP inflows trend upward across the observed period, with the supplied chart highlighting an all-time-high of ₹31,002 crore.
- Total folio count rises from 13.26 crore to 26.12 crore across January 2022 to December 2025.
- SBI Mutual Fund leads the ten displayed fund houses on AUM at the latest snapshot, at approximately ₹12.5 lakh crore.
- Small-cap and mid-cap schemes occupy the higher-return, higher-volatility region of the risk-return scatter; gilt and large-cap schemes occupy the lower-risk region.
- Banking, IT and Pharma are the largest sector exposures across the equity holdings analysed.
- The dashboard connects industry trends, fund performance, investor analytics and SIP/market trends in one SQLite-backed application; automation converts the analysis into a repeatable weekly reporting workflow.

22. Business Insights

For a data analyst, business analyst or investor-facing team, the platform's outputs translate into practical decision support:

- Fund selection support: the Fund Performance Scorecard (Section 18) gives a single ranked view combining return, risk-adjusted return, cost and drawdown, useful as a first-pass screen before deeper due diligence.
- Risk monitoring: VaR/CVaR and sector-HHI outputs (Sections 17–18) flag schemes with elevated downside risk or concentrated sector exposure, supporting portfolio-review conversations.
- Distribution strategy: investor demographic and city-tier findings (Section 15) can inform which investor segments a distributor or AMC marketing team may want to prioritise or better understand.
- Trend monitoring: the automated weekly report keeps AUM, SIP and folio KPIs current without manual rebuilding, supporting recurring management reporting.

- Category-level allocation view: monthly category inflow data (Section 16) shows where investor money is currently flowing, useful context for product and category strategy discussions.

23. Limitations

- Investor transaction data used for demographic and geographic analysis is synthetic and generated for educational analysis, not sourced from real investor records.
- Some figures in the supplied project materials are sourced or simulated/anchored for project purposes and should be interpreted within the supplied project-data context rather than as industry-wide statistics.
- The current Streamlit dashboard is a local application and does not automatically provide a public URL.
- Live NAV retrieval is implemented for selected schemes via mfapi.in; it should not be represented as a continuously live feed across all 40 schemes.
- Automated email delivery to external recipients depends on Resend account configuration and sender-domain verification; unrestricted production email delivery to arbitrary recipients is not claimed.
- The scheduled pipeline requires the host machine to be powered on; it does not run if the machine is fully powered off, unless Windows is separately configured to wake the machine for scheduled tasks.
- The project is local and educational in nature; it is not presented as a production financial platform, and no cloud or server deployment is claimed.
- The project's outputs are for analytics/educational use and do not constitute investment advice.

24. Future Enhancements

- Migrate from SQLite to PostgreSQL to support production-scale, multi-user operation.
- Deploy the dashboard publicly (e.g. Streamlit Community Cloud or a managed host) when a verified public URL is required.
- Broaden live-NAV coverage across the full 40-scheme universe with stronger API retry logic and monitoring.
- Add automated data-quality alerts and centralised logging for the ETL pipeline.
- Move scheduling from Windows Task Scheduler to a cloud scheduler with production-grade observability.
- Add authentication and role-based access control to the dashboard.
- Expand benchmark coverage and add portfolio-level attribution analytics.

25. Conclusion

Mutual Fund Analytics demonstrates a complete, practical workflow connecting data engineering, financial analysis, visualisation and automation. Across a validated 40-scheme universe, the project builds a repeatable pipeline from raw source data to a SQLite analytical layer, computes return, benchmark and risk metrics, surfaces the results through an interactive Streamlit dashboard, and automates weekly reporting and email delivery through Windows Task Scheduler.

The project demonstrates hands-on capability in Python and Pandas for data engineering, SQL/SQLite for analytical storage and querying, data-quality validation, financial performance and risk analysis, dashboard development, and Windows-based workflow automation — a practical skill set directly relevant to Data Analyst, Business Analyst and Associate Analyst roles in the FinTech domain.

26. Technology Stack

Layer	Technologies
Programming	Python
Data Analysis	Pandas, NumPy
Visualisation	Matplotlib, Seaborn, Plotly
Database	SQLite, SQL
Dashboard	Streamlit
Live Data	MFAPI (mfapi.in)
Automation	Python subprocess, Windows Task Scheduler
Email Reporting	Resend
Development Tools	Jupyter, VS Code

27. References

- AMFI India — public mutual-fund reference information and industry notes.
- MFAPI (mfapi.in) — public mutual-fund NAV API used by the project for selected live NAV retrieval.
- Supplied Bluestock Fintech capstone specification and project datasets.
- Project notebooks, SQL/database implementation, Streamlit dashboard code and official project visualisations supplied for this report.

Financial-data disclaimer: this document describes an educational analytics project. It does not constitute investment advice.